

Michael Phillips

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Security Clearance: Active DoD Secret

EDUCATION

Virginia Tech | Bachelor of Science in Computer Science

August 2022 - May 2026

EXPERIENCE

Technomics, Inc.

Associate | Arlington, VA

July 2026- Present

- Engineered automated Python data ingestion pipelines leveraging OCR and document-parsing algorithms to clean, structure, and process unstructured source documents for federal defense clients (DOE/NNSA).
- Architected relational database schemas and Entity-Relationship Diagrams (ERDs), designing dynamic querying logic and business rules to maintain cross-system data integrity.
- Integrated enterprise endpoints and RESTful data services to automate analytical workflows, reducing manual data processing overhead for client stakeholders.
- Collaborated directly with federal mission leads to translate ambiguous client requirements into deployed technical software specifications and analytical data pipelines.

The Math Tree

Co-Founder, Chief Technology Officer | Blacksburg, VA

May 2025- September 2025

- Co-founded and served as CTO, directing full-stack software architecture, cloud infrastructure, and technical operations for the web platform.
- Developed and maintained 5,000+ lines of production full-stack code using Git version control, delivering a market-ready MVP in under 2 months to attract venture capital interest.
- Engineered responsive UI/UX components, scalable REST API endpoints, and relational database schemas, supporting rapid adoption from 10 to 200+ active users within two weeks.
- Administered Microsoft Azure cloud infrastructure, provisioning App Services and database instances with automated deployment pipelines to ensure high availability.
- Led technical hiring, developer onboarding, and technical presentations to Virginia Tech staff and student organizations, aligning product development with user and business needs.

Peraton

Software Engineering Co-Op | Blacksburg, VA

May 2023- December 2023

- Developed responsive ReactJS/CSS interfaces and scalable back-end solutions using SQL databases and custom API endpoints for robust data management and seamless integration, translating client requirements into reliable full-stack features across devices and deployment environments.
- Collaborated with cross-functional teams and clients in Agile development using Jira to track requirements, prioritize work, and drive iterative releases, rapid problem resolution, testing, and continuous alignment with evolving technical and project goals while maintaining clear communication across stakeholders.

SOFTWARE ENGINEERING PROJECTS

AI-Assisted Statistical Trading and Risk Analysis Model

- Built Python data pipelines to collect, normalize, and analyze multi-timeframe market data using custom volatility, Bollinger Band, RSI, and risk indicators.
- Integrated local LLM agents to evaluate structured market signals and aggregate probability-based analytical outputs.
- Built a historical test harness to validate predictions against future price behavior and measure model performance.

Image Annotation Tool

- Developed a ReactJS image annotation platform supporting image uploads, unlimited bounding boxes, and XML export for computer vision datasets.
- Leveraged Redux for scalable state management and structured output optimized for high-volume machine learning datasets.

ASL Hand Sign Recognition Model

- Developed a transfer-learning and OpenCV model to recognize and classify American Sign Language (ASL) hand-signed letters from video input.
- Used custom labeled datasets from the annotation tool with preprocessing, augmentation, and evaluation workflows to improve recognition consistency.

SKILLS

- Languages:** Python, SQL, TypeScript, JavaScript, Java, C/C++, C#, Bash, PowerShell, HTML5/CSS3
- Frameworks & Libraries:** ReactJS, Redux, Node.js, OpenCV, Pandas, NumPy, Scikit-learn, REST APIs, Tesseract OCR
- Cloud, Data & Platforms:** Microsoft Azure (App Services, Database Servers), CI/CD, Git, GitHub/GitLab, Docker, Linux/Unix
- Enterprise & Data Platforms:** Microsoft Power Apps, Dataverse, Microsoft 365 Integration, Jira, Agile/Scrum, Relational Data Modeling (ERDs)